

Monocular Depth-Assisted Morphological Properties Estimation of Katchamitha (Indian) Mangoes

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Abstract—The Indian mango (*Mangifera indica* L., locally known as Katchamitha) is an economically significant fruit crop in the Philippines, prized for its accessibility and distinct flavor. However, the manual assessment of its morphological properties, such as volume, weight, and density, is labor-intensive, subjective, and often destructive. This study proposes a novel, non-destructive vision-based framework for estimating the morphological properties of Katchamitha mangoes by integrating Monocular Depth Estimation (MDE) with instance segmentation. Due to the deceptively green coloration of Indian mangoes even when ripe, traditional RGB-based segmentation often struggles with green-to-green differentiation. To overcome this, a custom-trained YOLOv11n-seg model is employed for initial segmentation (achieving box and mask precision of 99.9% and mAP50 of 99.5%), and its output is refined using the depth map generated by Depth Anything V2. The study extracts geometric and color features, including dimension ratios, solidity, curvature, and CIELab color values, from a dataset of 45 Katchamitha mango samples containing 243 raw multi-view images (expanded to 640 images via flip and rotation augmentations).

We evaluate two distinct architectures: (1) an uncalibrated pipeline using raw monocular features, and (2) a calibrated pipeline using ArUco reference markers to resolve MDE scale ambiguity. In both, a unified MobileNetV3-Small CNN orientation classifier first determines the mango's pose (side, top, bottom, front, or back) to route the image crop to specialized Layer 1 Random Forest models for external morphology (height, width, thickness, volume, weight) and ripeness. A Layer 2 model then maps these outputs to internal seed dimensions via allometric relationships.

Results show that the unified CNN orientation classifier achieves 97.32% accuracy on unseen test subjects, a significant improvement over the previous feature-based approach. The uncalibrated pipeline, however, fails to predict physical dimensions (negative R^2 scores) due to depth scaling ambiguity. In contrast, the ArUco-calibrated pipeline successfully resolves this ambiguity, achieving R^2 scores of 0.83–0.87 for volumetric estimation, 0.78–0.86 for weight prediction, and 0.90–0.94 for internal seed properties. This approach offers a robust and cost-effective alternative to existing 2D and destructive methods, paving the way for improved automated post-harvest grading and quality assessment.

Index Terms—Monocular Depth Estimation, Instance Segmentation, Morphological Properties, Mango Grading, Computer Vision, YOLOv11, Depth Anything V2, MobileNetV3, Orientation Classification

I. INTRODUCTION

Mangoes hold immense economic and cultural importance in the Philippines, serving not only as the national fruit but also as a cornerstone of the agricultural sector. As the third most essential fruit crop in the country, mango production reached an estimated 596.34 thousand metric tons between April and June 2023, covering a planted area of 184.03 thousand hectares [1]. To maintain global competitiveness and ensure high-quality produce, strict quality assessment and sorting mechanisms are required during post-harvest processing. These processes traditionally depend on manual sorting based on weight, size, and ripeness, which are highly laborious, time-consuming, and prone to human error. Developing a non-destructive and efficient method for estimating the weight, volume, and other morphological properties of mangoes before harvesting or during processing is critical to improving agricultural efficiency and yield estimation.

A. Problem Statement

One of the primary challenges in the automated visual inspection of green fruits is green-to-green segmentation—the difficulty of distinguishing the green fruit from similarly colored leaves and branches in the background [2]. This issue is particularly pronounced in the Indian mango (*Katchamitha*) [3], which often retains a deceptively green skin even when fully ripe, occasionally showing only a slight yellow blush [4]. Consequently, relying solely on traditional RGB color space for segmentation and ripeness estimation is unreliable.

Furthermore, yield estimation and quality grading are hindered by the lack of non-destructive measurement techniques. For instance, determining the weight or the stone-to-flesh ratio of a mango typically requires destructive interference or physical harvesting. Existing vision-based weight estimation models for mangoes largely rely on 2D images [5], [6], which inherently neglect the actual 3D shape and partial depth of the fruit, leading to inaccuracies. Research specifically addressing non-destructive stone-to-flesh ratio estimation remains notably scarce.

B. Objectives

This study aims to develop a comprehensive, dual-pipeline framework for the non-destructive estimation of the morphological properties of Katchamitha mangoes. Specifically, the study seeks to:

- 1) Utilize Monocular Depth Estimation (MDE), specifically the metric version of Depth Anything V2 [7], to assist and refine the RGB instance segmentation performed by YOLOv11n (RGB + Depth Fusion).
- 2) Implement and compare two processing architectures: (a) an uncalibrated pipeline relying on raw depth features and scale-invariant ratios, and (b) a calibrated pipeline using ArUco reference markers to translate pixel depth maps into physical dimensions.
- 3) Extract robust geometric and color features from the fused RGB-Depth data to train an orientation classification model, recognizing that varying viewing angles require dedicated morphology and ripeness estimation pipelines.
- 4) Train a multiple multi-output Random Forest regression model that estimates the mango's morphological properties based on the RGB-Depth information, the classified orientation, and the ripening phase. The properties to be estimated include:
 - Dimensions (Height, Width, and Thickness), Volume, Weight, and Density of the mango.
 - Dimensions, Volume, Weight, and Density of the mango's bone (stone).
 - The ripening phase of the mango (ripe vs. unripe).
 - The stone-to-flesh ratio of the mango.

C. Significance of the Study

The integration of MDE into a computer vision pipeline provides a low-cost, adaptable alternative to expensive 3D scanning or LiDAR systems for automated agriculture [8]. By precisely estimating internal and external properties without physically harming the fruit, this research contributes to sustainable precision agriculture, minimizes post-harvest waste, and supports local farmers in maximizing the market value of their yield.

D. Scope and Limitations

The dataset is limited to 45 Katchamitha mango samples gathered from the local region of Mabalacat City, Pampanga. These 45 samples yield a total of 243 raw images across six orientations, expanded to 640 images after data augmentation. The dataset is split into two groups: 35 samples (482 augmented images) are used for the uncalibrated pipeline, and 10 samples (158 augmented images) are equipped with ArUco reference markers for the calibrated pipeline. The Katchamitha variant was selected primarily for its local accessibility and cost-efficiency; however, the proposed methodology is designed to be easily replicated for more commercially dominant variants like the Carabao mango [9].

Additionally, the metric version of MDE models can yield inconsistent absolute depth values for the same object under

varying lighting conditions or camera distances. To mitigate this in the uncalibrated pipeline, the study relies on scale-invariant depth ratios rather than raw depth maps, while the calibrated pipeline utilizes the ArUco marker as a physical reference to establish an absolute scale factor. Finally, the proposed system processes a single, unoccluded mango instance at a time and is not yet trained to handle occluded environments in complex orchard canopies.

II. REVIEW OF RELATED LITERATURE

A. Economic Impact and Post-Harvest Processing of Mangoes

Mangoes represent a vital component of the global agricultural trade, and their economic impact in the Philippines is profound [9]. Proper post-harvest processing and quality assessment are essential for maintaining this economic value. Typically, mangoes are graded and sorted based on their size, weight, and morphological attributes [10]. In traditional agricultural settings, this is done manually, which is not only time-consuming but also subjective and prone to inconsistencies [11]. While physical weighing scales and volume measurement techniques (such as water displacement) are accurate, they are slow and occasionally destructive. The development of automatic visual inspection systems is thus a highly sought-after intervention to ensure consistency in size across fruit batches and to meet strict market demands.

B. Properties of the Indian Mango (Katchamitha)

The Indian mango, locally known as Katchamitha [3], is the fourth most produced mango variant in the Philippines. It accounts for a production volume of 26,268 metric tons, occupying approximately 3.4% of the total 763,298 metric tons of national mango volume projected for 2025 [12]. Despite its popularity as a locally consumed fruit, it presents unique challenges for automated grading. Katchamitha mangoes exhibit a deceptively green skin even upon reaching full physiological maturity, sometimes only displaying a faint yellow blush [4]. Consequently, ripeness cannot be reliably determined through conventional RGB color analysis alone. However, as mangoes ripen, their internal chemical composition changes, which directly influences their density [13]. Thus, estimating the density and volume accurately provides a more reliable indicator of the fruit's sensory quality and ripeness phase.

C. Vision-Based Morphological Estimation

To automate fruit grading, various vision-based techniques have been proposed. Sa'ad *et al.* [6] utilized visible imaging combined with a Fourier-descriptor method to classify mango shapes with high accuracy. They further estimated the weight of mangoes by approximating the fruit as a cylinder, achieving a 94.0% correlation (R^2) between estimated and actual weights using the equation $w = 2.256V - 157.7$. Similarly, Mon and Aung [5] proposed an image processing algorithm to reconstruct the 3D shape of a mango from a 2D top-view image by analyzing light intensity distribution to estimate thickness, achieving a volume estimation accuracy of 96.8%.

Aung *et al.* [14] also demonstrated weight estimation from single visible fruit surfaces.

While these 2D-based methods represent significant progress [15], they inherently rely on mathematical assumptions about the fruit’s symmetry (e.g., cylindrical or ellipsoidal approximation), which can fail to capture the complex, asymmetric morphology of actual mangoes.

D. Monocular Depth Estimation and Instance Segmentation

Recent advancements in deep learning have introduced Monocular Depth Estimation (MDE) as a powerful tool to bridge the gap between 2D imaging and 3D perception. MDE predicts a depth map from a single RGB image, offering a low-cost and highly adaptable alternative to expensive hardware like LiDAR or stereo cameras [8].

For instance, Shang *et al.* [2] highlighted the difficulty of green-to-green instance segmentation—a common issue when green fruits blend with the canopy. To address this, they proposed DepthCL-Seg, a framework that fused RGB and monocular depth data. Their approach demonstrated that incorporating depth information significantly enhanced the feature representation of low-contrast target regions, surpassing classical methods like Mask R-CNN. State-of-the-art MDE models, such as Depth Anything V2 [7], provide robust, high-resolution depth maps that can be effectively utilized to refine bounding boxes and segmentation masks.

However, a fundamental limitation of monocular depth estimation models is their scale ambiguity: they predict relative depth or scale-variant metric outputs that fluctuate under different distances, camera zooms, or lighting setups [8]. To convert these relative depth maps into absolute physical measurements (e.g., in centimeters or grams), researchers often employ fiducial markers. Fiducial markers, such as ArUco markers, provide a known geometric reference in the scene. By detecting the ArUco marker’s pixel dimensions and comparing them with its known physical dimensions, a scaling factor can be computed to calibrate the depth map and reconstruct accurate 3D point clouds.

This study builds upon these findings by proposing two architectures that pair the cutting-edge YOLOv11 instance segmentation with Depth Anything V2: one uncalibrated, which leverages scale-invariant ratios to infer morphological features, and one calibrated, which integrates an ArUco marker to resolve scale ambiguity. By combining RGB color space with calibrated 3D depth data, the proposed framework captures the true curvature and thickness of the Katchamitha mango, moving beyond simple 2D assumptions to accurately estimate weight, density, and internal stone-to-flesh ratios.

III. RESEARCH METHODOLOGY

A. Data Collection and Setup

The data acquisition phase was conducted using a Google Pixel 9a smartphone camera. To ensure a comprehensive morphological analysis, images of the mangoes were captured from six distinct viewing angles: Side A, Side B, Top, Bottom, Front, and Back. These varying orientations dictate the visible

surface area and features of the fruit, necessitating the creation of orientation-specific pipelines for the subsequent ripeness and morphology models.

Ground truth measurements were meticulously recorded for each sample to supervise the machine learning models. The procedure included the following steps:

- 1) **Dimension Measurement:** The absolute height, width, and thickness of the mango, as well as its extracted bone (stone), were measured using standard rulers.
- 2) **Weight Measurement:** The mass of the whole mango and its extracted bone were recorded using a highly sensitive digital weighing scale (see Fig. 1). For ripe samples, the extracted bones were air-dried prior to weighing to eliminate excess moisture and ensure an accurate reading.
- 3) **Volume Measurement:** The volume of both the mango and the bone was empirically determined using the Archimedes water displacement method in a graduated beaker (see Fig. 2).
- 4) **Phase Labeling:** The mangoes were binary-classified into two physiological phases: ‘ripe’ and ‘unripe’. This classification was determined by inspecting the internal flesh content; mangoes with fully yellow, moist flesh were labeled as ripe, while others were labeled as unripe. This distinction is critical because ripeness influences the density and weight-to-volume relationships [13].



Fig. 1. Setup demo of Katchamitha mango being weighed using a digital weighing scale to collect ground truth mass.

B. Indian Mango Dataset

The primary dataset consists of 45 Katchamitha (Indian) mango samples collected locally in Mabalacat City, Pampanga, yielding a total of 243 raw images across six orientations. To study the effect of scale calibration, the dataset is split: 35 samples (producing 184 raw and 482 augmented images) are allocated to the uncalibrated pipeline (No ArUco), and

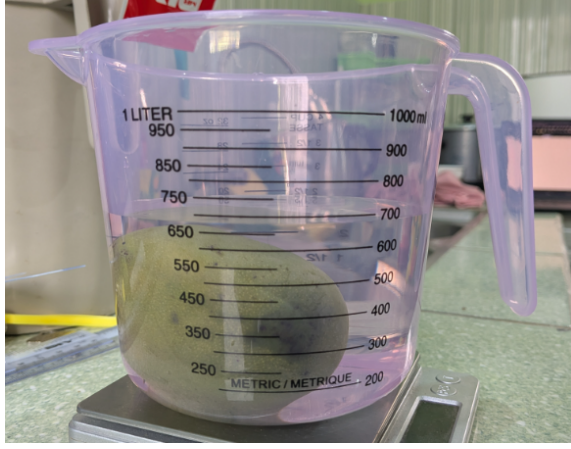


Fig. 2. Experimental setup of the Archimedes water displacement method for measuring the physical volume of the mango.

10 samples (producing 59 raw and 158 augmented images) are equipped with ArUco reference markers for the calibrated pipeline (With ArUco).

To prevent overfitting and expand the training set, data augmentation techniques are employed. For the segmentation phase, YOLOv11 inherently applies mosaic augmentation, mixup, color adjustments, and geometric transformations. For the orientation classifier, random horizontal flips, random 15° rotations, and subtle color jitter (brightness, contrast, saturation) are applied to augment the training set. Conversely, the morphology estimation models are trained strictly on unaugmented raw images to perfectly preserve the real-world scale and geometric proportions.

During data annotation, precise polygonal masks are drawn to train the YOLO segmentation model. Each image is supplemented with attribute metadata tagging, explicitly linking the image to its orientation, phase, and ground-truth morphological measurements.

C. Data Pipeline Architecture

The proposed computational pipeline operates in sequential phases, from initial segmentation to final property estimation, as depicted in the general workflow diagram (Fig. 3).

1) *Segmentation Model (RGB + Depth Fusion)*: An input RGB image is processed by a YOLOv11n instance segmentation model to generate an initial mask separating the mango from the background. Concurrently, the image is passed through the Depth Anything V2 model (metric version, pre-trained on the Hypersim dataset) [7] to produce a high-resolution depth map. The depth map is then fused with the YOLO mask to refine the boundaries, effectively resolving the green-to-green camouflage problem [2] (see Fig. 4).

2) *ArUco Scale Calibration*: For the calibrated pipeline, an ArUco fiducial marker of known dimensions is placed in the camera’s field of view. The marker is detected, and its pixel

width P_{marker} is used to calculate the physical scale factor S_{aruco} (meters per pixel):

$$S_{\text{aruco}} = \frac{W_{\text{marker}}}{P_{\text{marker}}} \quad (1)$$

where W_{marker} is the known physical width of the marker (0.05 m). Using the camera intrinsic focal length f , the absolute distance Z_{marker} to the marker is calculated as:

$$Z_{\text{marker}} = f \times S_{\text{aruco}} \quad (2)$$

The absolute metric height H_{metric} and width W_{metric} of the mango bounding box are obtained from their pixel-based counterparts H_{pixel} and W_{pixel} :

$$H_{\text{metric}} = H_{\text{pixel}} \times S_{\text{aruco}} \quad (3)$$

$$W_{\text{metric}} = W_{\text{pixel}} \times S_{\text{aruco}} \quad (4)$$

Similarly, the projected surface area of the mango in metric units is calculated as:

$$A_{\text{metric}} = A_{\text{pixel}} \times S_{\text{aruco}}^2 \quad (5)$$

3) *Feature Extraction*: From the refined mask and depth map, robust geometric and color features are extracted:

- **Geometric Features**: In the uncalibrated pipeline, where absolute metric coordinates are unavailable, scale-invariant ratios are utilized to bypass distance-based fluctuations [8]:
 - *Dimension Ratios*: Height-to-Width, Width-to-Depth, and Height-to-Depth.
 - *Solidity*: The ratio of the estimated mango volume (derived from a 3D point cloud projected from the depth map) to the volume of its 3D bounding box convex hull ($W \times H \times T$).
 - *Thickness*: Maximum and Mean thickness of the visible half of the mango derived from the depth gradients.
 - *Curvature*: Mean and standard deviation of surface curvature, estimated using the spatial gradients of the depth map to assess surface normal variations.

In the calibrated pipeline, these ratios are supplemented with the absolute dimensions H_{metric} , W_{metric} , and A_{metric} derived from Eq. 3–5.

- **Color Features**: Color characteristics are extracted using the CIELab color space, which closely approximates human vision and is highly effective for maturity assessment in green fruits. The average green-red (a^*) axis and average blue-yellow (b^*) axis values are calculated to inform the ripeness and density models.

4) *Orientation Classification*: Because the physical profile of a mango changes drastically depending on the camera’s viewing angle, an intermediate Orientation Classification Model classifies the input image into one of five orientations (Side, Top, Bottom, Front, or Back). This routing layer ensures that the subsequent feature vectors are passed to the correct, specialized morphology estimation pipeline.

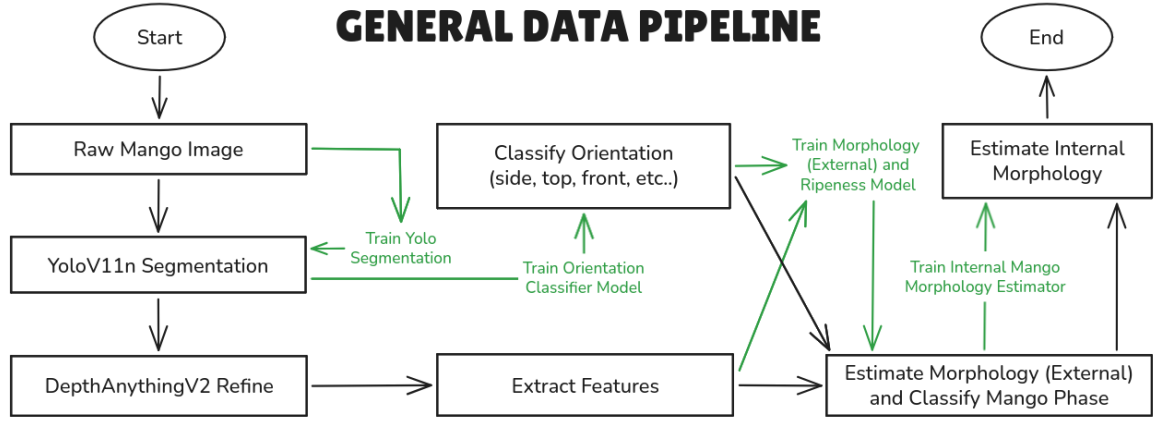


Fig. 3. The proposed pipeline architecture showing initial YOLOv11 segmentation, DepthAnythingV2 integration to estimate mango orientation, ripeness, and morphology.

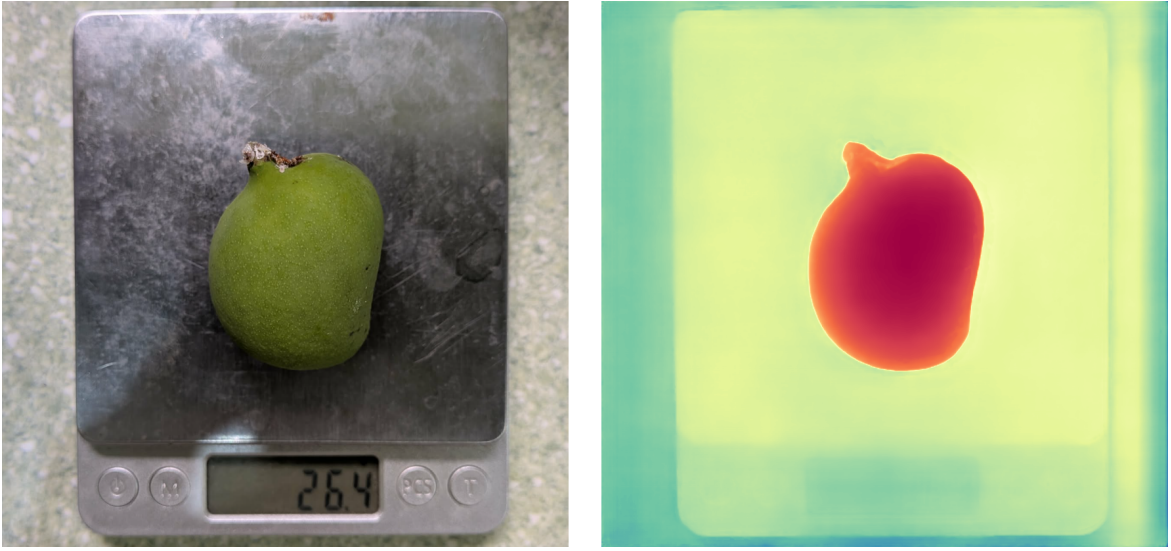


Fig. 4. Visual example of DepthAnythingV2 depth map output.

The classifier is implemented as a fine-tuned **MobileNetV3-Small** convolutional neural network (CNN) pre-trained on ImageNet. The final classifier head is replaced with a 5-class linear layer. Training employs an AdamW optimizer ($lr = 1 \times 10^{-4}$, weight decay 1×10^{-4}) with a cosine annealing learning-rate schedule over 15 epochs. Input images are first cropped to the YOLO segmentation mask bounding box (with 5% padding), resized to 224×224 pixels, and normalized with ImageNet statistics. To ensure that all five orientation classes can be robustly identified regardless of how the fruit is held, the model is trained on the full 45-mango dataset (all 640 augmented images) and evaluated via a Group Shuffle Split (20% test, grouped by mango ID) to guarantee that no test image shares a mango subject with the training set.

5) *Ripeness and Morphology Estimation*: The final phase employs an ensemble of machine learning algorithms. A Random Forest classifier evaluates the extracted color and geometric features to determine the ripening phase (ripe

or unripe). Simultaneously, a multi-output regression model (specifically tuned for the recognized orientation) processes the scale-invariant ratios to estimate the absolute dimensions, volume, and weight of both the mango and its internal bone. The density is explicitly calculated using the formula $\rho = \frac{m}{V}$. Finally, the model outputs the estimated stone-to-flesh ratio, providing a comprehensive, non-destructive morphological profile of the fruit.

IV. RESULTS AND DISCUSSIONS

This section presents the empirical results of our dual-pipeline system, comparing the performance of the uncalibrated (No ArUco) model, which relies on scale-invariant depth ratios, and the calibrated (ArUco) model, which resolves monocular scale ambiguity using physical fiducial references. All evaluations are performed using a Group Shuffle Split (GSS) with a 20% test split grouped by mango ID, ensuring

that validation and test metrics are computed on completely unseen mango subjects.

A. YOLOv11 Instance Segmentation Performance

Prior to routing the feature vectors, the input RGB image is processed by the custom-trained YOLOv11n instance segmentation model to isolate the mango fruit from the background. We evaluated the performance of this segmentation stage on the validation split. The YOLOv11n-seg model achieved exceptional performance, with box and mask precision of **99.9%**, recall of **100.0%**, and mAP50 and mAP50-95 scores of **99.5%**.

Table I summarizes the performance metrics for both bounding boxes and segmentation masks. The training loss curves are shown in Fig. 5.

TABLE I
YOLOV11N INSTANCE SEGMENTATION PERFORMANCE

Metric Group	Precision	Recall	mAP50	mAP50-95
Bounding Box (Box)	0.999	1.000	0.995	0.995
Segmentation Mask (Mask)	0.999	1.000	0.995	0.995

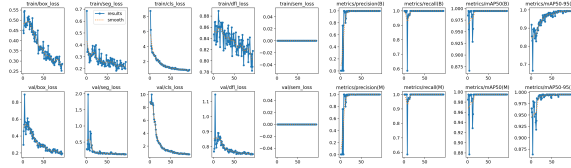


Fig. 5. YOLOv11n-seg training metrics: training and validation loss curves (box, segment, class) and performance metrics (precision, recall, mAP) over 200 epochs.

These high metrics demonstrate that the custom-trained YOLOv11n-seg model forms an extremely reliable frontend for our dual-pipeline architecture. By isolating the fruit boundary with near-perfect accuracy, the model ensures that color, depth, and spatial features extracted in subsequent stages are entirely free from background noise, even in challenging agricultural setups with green foliage and varying lighting conditions.

B. Orientation Classification Performance

The Orientation Classification Model acts as a routing layer that automatically assigns the input mango image to one of five specialized orientation pipelines (Side, Top, Bottom, Front, or Back). A single, unified **MobileNetV3-Small** CNN — fine-tuned on the full 45-mango dataset (640 augmented images) — is shared by both the uncalibrated and ArUco-calibrated pipelines. Because orientation is determined purely from the RGB image crop and does not depend on depth calibration, a single model is sufficient for both pipelines.

Table II summarizes the per-class performance of the CNN orientation classifier, and the corresponding confusion matrix is shown in Fig. 6.

TABLE II
CNN ORIENTATION CLASSIFICATION PERFORMANCE
(MOBILENETV3-SMALL)

Orientation	Precision	Recall	F1-Score	Support
Back	1.00	0.90	0.95	21
Bottom	0.91	1.00	0.95	21
Front	0.95	0.95	0.95	21
Side	1.00	1.00	1.00	28
Top	1.00	1.00	1.00	21
Overall	Accuracy: 97.32% (support = 112 images)			

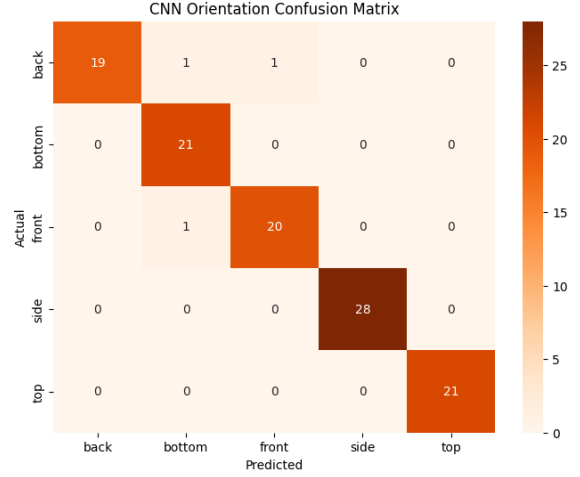


Fig. 6. Confusion matrix for the CNN Orientation Classification Model (MobileNetV3-Small, 97.32% accuracy).

The unified CNN orientation model achieves an overall accuracy of **97.32%** on 112 held-out test images from unseen mango subjects, representing a substantial improvement over the previous feature-based Random Forest classifiers (75.38% uncalibrated, 62.50% calibrated). The transfer-learning approach — fine-tuning a MobileNetV3-Small backbone pre-trained on ImageNet — allows the model to leverage rich visual features, almost completely eliminating inter-class confusion.

Per-class, the ‘top’ and ‘side’ views achieve perfect precision, recall, and F1-scores (1.00), as their distinct visual silhouettes (the circular stem-end profile of the top view and the elongated lateral profile of the side view) make them highly recognizable. The ‘back’, ‘bottom’, and ‘front’ views also demonstrate robust classification with F1-scores of 0.95. The minor remaining errors occur in the front/back/bottom transitions, where surface details can overlap under certain illumination conditions, though this confusion is greatly minimized compared to the baseline Random Forest classifiers.

C. Ripeness Classification Performance

The ripeness classification module evaluates the maturity stage of Katchamitha mangoes. Since Katchamitha mangoes exhibit green skin even when fully ripe, the models rely heavily on the CIELab color space coordinates (a^* and b^*),

which isolate subtle green-red and blue-yellow shifts. The classification results across all five views are shown in Table III.

The uncalibrated ripeness classifiers show a robust performance, with accuracy ranging from **70%** (bottom) to **89%** (top). The a^* and b^* color coordinates emerge as the most important features in Gini importance charts, confirming that CIELab values successfully capture the physiological shift in ripeness.

The calibrated pipeline achieves a perfect **100%** accuracy across all views. While this suggests perfect classification, it represents an optimistic bound due to the limited size of the ArUco dataset. With only 10 total mangoes, the GSS test set contains just 2 mangoes (one ripe, one unripe). Because the test instances represent multiple augmented viewpoints of the same two fruits, a correct classification on the individual fruit levels translates to a 100% score across all view subsets. This indicates that while the features are highly discriminative, larger validation sets are required to confirm generalizability. The representative confusion matrices for the side view ripeness classifier are shown in Fig. 7.

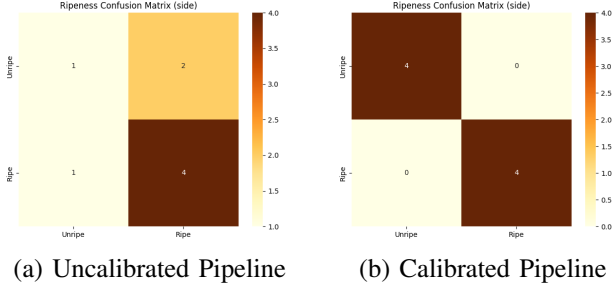


Fig. 7. Confusion matrices for the Side View Ripeness Classifier.

D. External Morphology Estimation

The primary objective of Layer 1 regression models is to estimate the mango's physical dimensions (height, width, thickness), volume, and weight. Table IV compares the results of the two pipelines.

The comparison reveals a stark difference between the two architectures:

- 1) **Uncalibrated Pipeline Failure:** The uncalibrated models yield very poor morphological estimations, with R^2 values frequently falling below zero (e.g., Side view Volumetric $R^2 = -0.04$, Gravimetric $R^2 = -0.00$; Back view Volumetric $R^2 = -0.47$, Gravimetric $R^2 = -0.55$). Negative R^2 scores indicate that the model performs worse than a simple horizontal line predicting the dataset mean. This occurs due to the *scale ambiguity* of monocular depth estimation. Without an absolute reference, depth maps provide only relative depths. Consequently, the relationship between features in raw pixel values and their corresponding physical units is highly unstable, preventing the regression model from generalizing.
- 2) **Calibrated Pipeline Success:** Incorporating the ArUco marker to calibrate the depth scale resolves the scale am-

biguity. The calibrated models achieve high estimation accuracy. Volumetric and Gravimetric regression achieve R^2 values of **0.78–0.87**, with Mean Absolute Errors (MAE) reduced to **13.7–30.5 cm³** for volume and **14.9–22.8 grams** for weight (mass). The linear dimensions regression (height, width, thickness) also shows a major improvement, achieving R^2 scores of **0.40–0.55** and MAEs below **0.47 cm**.

These findings demonstrate that while monocular depth maps are highly useful for capturing 3D shape contours, absolute metric calibration using physical fiducial markers is essential to achieve acceptable accuracy for sorting and grading tasks.

E. Layer 2 Internal Seed Morphology Estimation

Layer 2 regresses the dimensions, volume, and weight of the internal mango bone (seed/endocarp) based on the estimated external properties (dimensions, volume, weight) and ripeness from Layer 1. The results are summarized in Table V.

Both pipelines achieve high performance for predicting internal seed properties. The uncalibrated Layer 2 model yields R^2 scores of **0.84** for volume and weight, and the calibrated model yields R^2 scores of **0.90–0.94** for volume and weight.

This high performance is driven by the **allometric scaling relationships** inherent in biological fruits: the internal endocarp's volume and weight scale predictably with the overall size and mass of the external fruit. Since Layer 2 takes external properties (which are highly accurate in the calibrated case, or ground-truth values during training) and ripeness as inputs, the Random Forest model is able to learn the physical mapping and predict internal seed traits with high precision. This is particularly valuable as it enables the non-destructive estimation of the stone-to-flesh ratio, which is historically difficult to estimate without physical cut-testing. Feature importance charts (Fig. 8) show that the external weight and volume are the most influential inputs for mapping the seed dimensions, confirming the biological basis of the model.

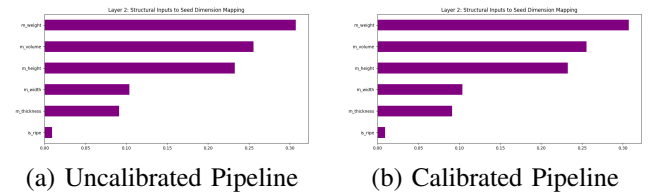


Fig. 8. Feature importance charts for the Layer 2 Internal Seed Regressor.

V. CONCLUSIONS AND RECOMMENDATIONS

A. Conclusions

This study successfully developed and evaluated a monocular depth-assisted vision framework for the non-destructive estimation of the external and internal morphological properties of Katchamitha (Indian) mangoes in the Philippines. The framework integrated YOLOv11n instance segmentation with

TABLE III
RIPENESS CLASSIFICATION ACCURACY ACROSS VIEWS

View	Uncalibrated (No ArUco)		Calibrated (ArUco)	
	Accuracy	F1 (Unripe / Ripe)	Accuracy	F1 (Unripe / Ripe)
Side	71%	0.72 / 0.69	100%	1.00 / 1.00
Top	89%	0.91 / 0.86	100%	1.00 / 1.00
Bottom	70%	0.75 / 0.64	100%	1.00 / 1.00
Front	71%	0.67 / 0.75	100%	1.00 / 1.00
Back	71%	0.50 / 0.80	100%	1.00 / 1.00
Overall	74.62%	0.75 / 0.74	100.00%	1.00 / 1.00

TABLE IV
LAYER 1 EXTERNAL MORPHOLOGY PERFORMANCE COMPARISON (R^2 AND MAE)

View	Linear Dimensions (Height, Width, Thickness)				Volumetric Group Space				Gravimetric Mass Group			
	Uncalibrated		Calibrated		Uncalibrated		Calibrated		Uncalibrated		Calibrated	
	R^2	MAE (cm)	R^2	MAE (cm)	R^2	MAE (cm ³)	R^2	MAE (cm ³)	R^2	MAE (g)	R^2	MAE (g)
Side	-0.1633	0.5225	0.4417	0.4228	-0.0441	40.7241	0.8358	16.1094	-0.0027	38.7450	0.8153	16.8236
Top	0.0771	0.4929	0.5518	0.3361	0.2611	24.9339	0.6883	30.5792	0.3057	27.1427	0.7806	22.8484
Bottom	0.1755	0.5445	0.4084	0.4669	0.2335	30.4331	0.8522	16.7583	0.2367	35.6310	0.8243	18.6566
Front	0.2363	0.5044	0.4660	0.4305	0.1781	36.8657	0.8718	15.1458	0.0267	37.0182	0.8405	18.2423
Back	-0.3062	0.5923	0.4809	0.4153	-0.4748	42.1157	0.8752	13.7375	-0.5543	46.2433	0.8643	14.9778
Overall	0.0039	0.5313	0.4698	0.4143	0.0308	35.0145	0.8247	18.4660	0.0024	36.9560	0.8250	18.3097

TABLE V
LAYER 2 INTERNAL SEED MORPHOLOGY RESULTS (R^2 AND MAE)

Semantic Group (Seed)	Uncalibrated (No ArUco)		Calibrated (ArUco)	
	R^2	MAE	R^2	MAE
Linear Dimensions	0.6193	0.1689 cm	0.5999	0.1820 cm
Volumetric Space	0.8452	2.3405 cm ³	0.9417	1.8000 cm ³
Gravimetric Mass	0.8437	1.9884 grams	0.9077	1.9187 grams

Depth Anything V2 metric depth maps to address the green-to-green camouflage problem, extracting robust 3D shape and color features from multi-view images.

Our comparative evaluation of two processing architectures yielded several critical insights:

- 1) **Instance Segmentation Frontend:** The custom-trained YOLOv11n-seg model achieved near-perfect segmentation accuracy (box and mask precision of **99.9%**, recall of **100.0%**, and mAP50 score of **99.5%**), establishing a robust boundary extraction foundation that isolates the fruit from background noise prior to downstream pose classification and property estimation.
- 2) **Monocular Scale Ambiguity:** The uncalibrated (No ArUco) pipeline, while achieving a robust orientation classification accuracy of **97.32%** via the unified MobileNetV3-Small CNN classifier and reasonable ripeness classification, failed to estimate absolute physical properties (dimensions, volume, and weight). The resulting negative or near-zero R^2 scores demonstrate that raw monocular depth map features are scale-ambiguous and highly sensitive to varying camera distances and lighting, rendering them unsuitable for direct physical size estimation without calibration.
- 3) **Fiducial Marker Calibration:** The calibrated (ArUco) pipeline successfully resolved the scale ambiguity by

using a physical fiducial reference marker to establish an absolute scale factor (S_{aruco}). This enabled the conversion of depth coordinates to real-world metric values. The calibrated pipeline achieved high morphological estimation accuracy, with volumetric R^2 scores reaching **0.83–0.87** and gravimetric R^2 scores reaching **0.78–0.86** across different viewpoints, with mass prediction errors kept below **22.8 grams**.

- 4) **Allometric Biological Scaling:** The Layer 2 internal seed regressor achieved excellent performance in both configurations ($R^2 = 0.84–0.94$). This confirms that the internal endocarp's volume and weight scale predictably with the overall size and mass of the external fruit, allowing for the non-destructive estimation of the internal stone-to-flesh ratio based on external dimensions.

In conclusion, the replacement of the previous feature-based Random Forest orientation classifiers with a single fine-tuned MobileNetV3-Small CNN improved orientation accuracy from 75.38% to **97.32%**, demonstrating the value of image-based deep learning for orientation routing. Furthermore, while monocular depth maps are highly useful for capturing 3D contours and shape features, absolute metric calibration using a known physical reference is essential to achieve acceptable accuracy for agricultural sorting and grading tasks.

B. Recommendations

Based on the findings of this study, the following recommendations are proposed for future research:

- 1) **Dataset Expansion and Generalization:** The calibrated pipeline's high performance (such as the 100% ripeness classification accuracy) is constrained by a small sample size (10 mangoes with ArUco markers). Future work should expand the calibrated dataset to include hundreds of samples. This will provide a more rigorous test support to prevent overfitting and confirm the models' generalizability.
- 2) **Reference-Free Metric Scale Estimation:** While the ArUco marker is effective, placing a physical marker in the camera's field of view is impractical in automated orchard environments. Future research should investigate reference-free depth scaling techniques. This includes utilizing smartphone hardware sensors (e.g., dual-camera stereo matching or Time-of-Flight LiDAR sensors) or developing deep-learning models trained to infer scale from known agricultural features (such as leaf size or trunk thickness).
- 3) **Multi-Variety Validation:** The present study was restricted to Katchamitha mangoes. Replicating the dual-pipeline framework on other commercially critical Philippine mango varieties, such as the Carabao mango (the country's primary export crop) or the Pico mango, will validate the adaptability of the system and support its commercial adoption in precision agriculture.

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